

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12408258
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Zhou; Zilu et al.

Hot cathode for an ion source of a neutron generator

Abstract

Systems and methods are provided for operating and constructing a neutron generator. In some examples, an ion source within a neutron generator may include a hot cathode; at least one cylindrical sleeve coupled to an exterior surface of the hot cathode; at least one electrode coupled to an interior surface of the neutron generator; and at least one mounting flange, wherein a proximal end of the at least one mounting flange is coupled to the at least one cylindrical sleeve and a distal end of the at least one mounting flange is coupled to the at least one electrode.

Inventors: Zhou; Zilu (Needham, MA), Guo; Weijun (Houston, TX), Fessler; Lesley Allison (Spring, TX)

Applicant: Halliburton Energy Services, Inc. (Houston, TX)

Family ID: 93379874

Assignee: HALLIBURTON ENERGY SERVICES, INC. (Houston, TX)

Appl. No.: 18/144500

Filed: May 08, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240381521 A1	Nov. 14, 2024

Publication Classification

Int. Cl.: H05H3/06 (20060101)

U.S. Cl.:

CPC H05H3/06 (20130101);

Field of Classification Search

CPC: H05H (3/06); H05H (6/00); G01V (5/105); H01J (35/045); H01J (35/04); H01J (2235/081); H01J (2235/06)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4075526	12/1977	Grubis	N/A	N/A
5293410	12/1993	Chen et al.	N/A	N/A
9322262	12/2015	Chirovsky et al.	N/A	N/A
10502861	12/2018	Navarro-Sorroche et al.	N/A	N/A
2002/0166975	12/2001	Reyes	250/423R	H01J 37/08
2009/0146052	12/2008	Groves	250/269.4	H05H 3/06
2011/0249801	12/2010	Firestone et al.	N/A	N/A
2013/0180780	12/2012	Chirovsky et al.	N/A	N/A
2014/0086376	12/2013	Perkins	376/114	H05H 3/06
2018/0090297	12/2017	Holden	N/A	H01J 37/08
2019/0304736	12/2018	Matsuura et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
1100110	12/2003	EP	N/A

OTHER PUBLICATIONS

International Search Report & Written Opinion; PCT Application No. PCT/US2023/021533; mailed Jan. 25, 2024. cited by applicant

Abstract of Ellis, Darwin et al., “Well Logging for Earth Scientists”, Springer Science & Business Media, Jun. 29, 2007—Science—708 pages. cited by applicant

Abstract of Csikai, G J., “CRC handbook of fast neutron generators”, United States: N. p., Jan. 1, 1987. cited by applicant

Abstract of Reifenschweiler, O., “Sealed-Off Neutron Tube: the Underlying Research Work”, Journal Philips Res. Rept., vol. 16; Oct. 1, 1961. cited by applicant

Schlumberger; Litho Scanner, High-definition spectroscopy service brochure; retrieved from <https://www.slb.com/-/media/files/fe/brochure/litho-scanner-br.ashx> on May 8, 2023. cited by applicant

Schlumberger; Pulsar Multifunction spectroscopy service brochure; retrieved from <https://www.slb.com/-/media/files/fe/brochure/pulsar-spectroscopy-br.ashx> on May 8, 2023. cited by applicant

Halliburton; RMT-3D™ Reservoir Monitor Tool for Oil, Water, and Gas Saturation; retrieved from <https://www.halliburton.com/en/products/reservoir-monitor-tool-3-detector> on May 8, 2023. cited by applicant

Abstract of Chen, Jianxing et al., “A new Cased-Hole 2½-in. Multi-Detector Pulsed-Neutron Tool: Theory and Characterization”, Paper presented at the SPWLA 56th Annual Logging Symposium, Long Beach, California, USA, Jul. 2015; SPWLA-2015-AAA; Jul. 18, 2015. cited by applicant

Mott, N.F., “The Theory Of Atomic Collisions”, Digital Library of India Item 2015.3748; 1949. cited by applicant

Primary Examiner: Porta; David P

Assistant Examiner: Faye; Mamadou

Attorney, Agent or Firm: NOVAK DRUCE CARROLL LLP

Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates generally to wellbore operations and, more specifically (although not necessarily exclusively), to a hot cathode for an ion source of a neutron generator the is operable in a wellbore environment.

BACKGROUND

(2) Wells can be drilled to access and produce hydrocarbons such as oil and gas from subterranean geological formations. Wellbore operations can include drilling operations, completion operations, fracturing operations, and production operations. Drilling operations may involve gathering information related to downhole geological formations of the wellbore. The information may be collected by wireline logging, logging while drilling (LWD), measurement while drilling (MWD), drill pipe conveyed logging, or coil tubing conveyed logging.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The various advantages and features of the present technology will become apparent by reference to specific implementations illustrated in the appended drawings. A person of ordinary skill in the art will understand that these drawings only show some examples of the present technology and would not limit the scope of the present technology to these examples. Furthermore, the skilled artisan will appreciate the principles of the present technology as described and explained with additional specificity and detail through the use of the accompanying drawings in which:

(2) FIG. 1 is a diagram of a wellbore system with a logging tool having an ion source for a neutron generator, in accordance with aspects of the present disclosure;

(3) FIG. 2 is a diagram of a neutron generator that includes an ion source and a target, in accordance with aspects of the present disclosure;

(4) FIG. 3 is diagram of a neutron generator that includes an ion source with a hot cathode and a target, in accordance with aspects of the present disclosure;

(5) FIG. 4A is a diagram of a hot cathode assembly for use in an ion source of a neutron generator, in accordance with aspects of the present disclosure;

(6) FIG. 4B is another diagram of a hot cathode assembly for use in an ion source of a neutron generator, in accordance with aspects of the present disclosure;

(7) FIG. 5A is a diagram of a hot cathode assembly configured within an ion source, in accordance with aspects of the present disclosure;

(8) FIG. 5B is another diagram of a hot cathode assembly configured within an ion source, in

accordance with aspects of the present disclosure;

(9) FIG. 6A is a diagram illustrating construction of a neutron generator that includes a hot cathode assembly, in accordance with aspects of the present disclosure;

(10) FIG. 6B is another diagram illustrating construction of a neutron generator that includes a hot cathode assembly, in accordance with aspects of the present disclosure;

(11) FIG. 7 is a graph illustrating electron emission current as a function of grid voltage for a hot cathode of a neutron generator, in accordance with aspects of the present disclosure;

(12) FIG. 8 is a flowchart diagram of a process for constructing a neutron generator, in accordance with aspects of the present disclosure; and

(13) FIG. 9 is a block diagram illustrating an example computing device architecture, in accordance with aspects of the present disclosure.

DETAILED DESCRIPTION

(14) The detailed description set forth below is intended as a description of various configurations of the subject technology and is not intended to represent the only configurations in which the subject technology can be practiced. The appended drawings are incorporated herein and constitute a part of the detailed description. The detailed description includes specific details for the purpose of providing a more thorough understanding of the subject technology. However, it will be clear and apparent that the subject technology is not limited to the specific details set forth herein and may be practiced without these details. In some instances, structures and components are shown in block diagram form in order to avoid obscuring the concepts of the subject technology.

(15) As discussed previously, wellbore drilling and/or operation can involve gathering information related to downhole geological formations of the wellbore. In some cases, a compact deuterium (D) and tritium (T) neutron generator can be used in a downhole nuclear logging tool for oil and gas well measurements. The neutron generator can include a sealed tube as vacuum housing, a gas reservoir for storing D/T gas, an ion source for generating ions that are accelerated by a high voltage system, and a target for facilitating the DT fusion reactions to generate neutrons.

(16) Many neutron generating tubes are based on hot-cathode ion source technology, in which a dispenser (hot cathode) is introduced to emit electrons for direct ionization of gas molecules to produce the D/T ions. In some cases, ions can be continuously extracted in a continuous wave (CW) or a pulsed mode, and the ions can be accelerated to bombard a target that contains D/T molecules which is powered at a high voltage (e.g., 100 kV).

(17) However, there are challenges associated with constructing and operating a hot cathode ion source for a neutron generating tube. For example, complex tube structure and compact geometry can make the hot cathode installation difficult and tedious. In some aspects, the tube structure with housing and electrodes may be brazed together in a high temperature environment, such that the hot-cathode and/or other delicate components can only be mounted after the brazing process. In some cases, an extended installation procedure can be problematic because the cathode surface may absorb moisture if it is exposed to air for an extended period of time, which can degrade its performance. Further, the performance of the neutron generator can be affected if the position of the hot cathode is incorrect or misaligned. In addition, it is important to minimize thermal losses during operation of the hot cathode in order to increase efficiency and performance.

(18) The disclosed technology addresses the foregoing by providing systems and techniques for implementing a hot cathode for use in an ion source of a neutron generator. For example, according to aspects of the present technology, a hot cathode can include one or more cylindrical sleeves that can be used for mounting the hot cathode within a neutron generator. In some cases, the cylindrical sleeves may include a thermal shield sleeve that can help minimize thermal losses.

(19) In some aspects, the hot cathode assembly disclosed herein can include a mounting flange that can be coupled to an electrode within the neutron generator in a manner that can precisely control the distance between the hot cathode and a corresponding grid. In some instances, the mounting flange can be centralized (e.g., within the tube) by a curb structure on the hot cathode electrode in

the tube to enhance positional accuracy.

(20) In some instances, the hot cathode assembly disclosed herein can include an electrical field (e-field) shaping structure that is disposed on the hot cathode electrode (e.g., directed toward the corresponding grid). In some examples, the hot cathode assembly disclosed herein can include through holes to facilitate vacuum connectivity.

(21) FIG. 1 is a diagram of a wellbore system **100** with a logging tool having an ion source that includes a hot cathode and is used for a neutron generator, according to some aspects of the present disclosure. In some cases, the wellbore system **100** may include a well, such as an oil or gas well, for extracting fluids from a subterranean formation **101**. In some examples, the wellbore system **100** may be used to create a wellbore **102** from a surface **110** of the subterranean formation **101**. In some aspects, the wellbore system **100** can include a well tool or downhole tool **118** and a drill bit **120**. In some configurations, the downhole tool **118** can be any tool used to gather information about the wellbore **102**. For example, the downhole tool **118** can be a tool delivered downhole by wireline, often referred to as wireline formation testing (“WFT”). In another example, the downhole tool **118** can be a tool for measuring-while-drilling, for measuring while wireline logging, and/or for measuring while logging-while-drilling.

(22) In some aspects, the downhole tool **118** can include a neutron generator **131** and one or more measurement devices such as measurement device **132** that may be used for determining information about the formation **101** through the wellbore **102**. For example, neutron generator **131** and/or measurement device **132** can be used to determine formation porosity, hydrogen index, sigma, density, elemental weight percentages, rock types, fluidic types (e.g., gas, oil, or water), and/or fluidic information in the wellbore **102**.

(23) In some aspects, the downhole tool **118** can include an additional sensor component **122** for determining information about the wellbore **102**. Examples of information can include monitoring of drilling rate of penetration, weight on bit, standpipe pressure, depth, mud flow in, rotations per minute, torque, equivalent circulation density, or other parameters. In some instances, the downhole tool **118** can also include a transmitter **124** for transmitting data from the sensor component **122** to the surface **110**. In some cases, the downhole tool **118** can include the drill bit **120** for drilling the wellbore **102**.

(24) In some examples, the wellbore **102** can be drilled from the surface **110** and through the subterranean formation **101**. In some aspects, drilling fluid can be pumped through the drill bit **120** and into the wellbore **102** to enhance drilling operations (e.g., as the wellbore **102** is drilled). In some cases, the drilling fluid can circulate back toward the surface **110** through a wellbore annulus **128**.

(25) In some cases, wellbore system **100** can also include a computing device **126**. In some examples, the computing device **126** can be communicatively coupled to the downhole tool **118** and receive data about the drilling or logging process. In some instances, the computing device **126** can process and display the data to a user. In some aspects, the computing device **126** include one or more of the components as illustrated in computing device architecture **900**.

(26) In some aspects, a compact deuterium (D) and tritium (T) neutron generator can be used in a downhole nuclear logging tool for oil or gas well measurements. For instance, the neutron generator can include a sealed tube as vacuum housing, a gas reservoir for storing gas (e.g., D-T gas), an ion source for generating ions that are accelerated by a high voltage system, and a target for facilitating the D-T fusion reactions to generate neutrons.

(27) FIG. 2 is a diagram of a neutron generator **200** that includes an ion source **204** and a target (e.g., film target **210** and target rod **212**), in accordance with aspects of the present disclosure. In some cases, neutron generator **200** can include a housing **202** that can be used to form a sealed tube. In some examples, the housing **202** of neutron generator **200** can include one or more ceramic housing rings (e.g., used as insulating spacers) and one or more metal washers (e.g., used as electrodes **224**) in a brazed stack.

(28) In some aspects, neutron generator **200** can be filled with a mixture of D.sub.2 and T.sub.2 gas, which can be stored in gas reservoir **208**. In some cases, neutron generator **200** can include ion source **204** that can use the gas stored in gas reservoir **208** to generate ion beam **222** (further details regarding ion source **204** are discussed below). In some examples, ion beam **222** can be directed at a target that can include film target **210** and target rod **212**. In some cases, film target **210** can be coated onto or otherwise attached to target rod **212**. In some cases, target rod **212** can be made of copper and may be used as a backing structure. In some configurations, film target **210** can include a Titanium layer that is saturated with the D-T gas (e.g., same gas stored in gas reservoir **208**).

(29) In some instances, the target rod **212** can act as an electrical connector to a high voltage (HV) source **220** and a thermal conductor to transfer heat away from the target. That is, D and T atomic and molecular ions can be generated by the ion source **204** and can be accelerated to bombard the film target **210**, which is loaded with the same gas. In some aspects, the D-T, or T-D fusion reactions can occur at a high voltage (e.g., supplied by HV source **220**) in order to generate neutrons.

(30) In some cases, neutron generator **200** may also include a resistor **218** that can be connected between the target rod **212** and the HV source **220**. In some aspects, the HV source **220** may be coupled to a corona shield **216** that can connect to a suppressor **214**. For example, corona shield **216** can be coupled outside of housing **202** and provide a connection to suppressor **214**. In some configurations, the suppressor **214** can be configured to reject or suppress low-energy, secondary emission electrons emitted from film target **210** during ion beam **222** bombardment. In further examples, the suppressor **214** can trap ions that reflect or scatter from film target **210** (e.g., backscattered ions). That is, the suppressor **214** may surround or enclose film target **210** and a portion of target rod **212** to trap backscattered ions and to suppress secondary emission electrons within the housing **202** of neutron generator **200**.

(31) In some examples, neutron generator **200** can include a vacuum seal **206** (e.g., an end cap structure) that can be used to seal the vacuum enclosure. In some cases, the vacuum seal **206** may include a tubing structure (e.g., copper tubing) that can be connected to a vacuum pumping and gas handling system for neutron tube processing. In some aspects, the tubing structure that is part of vacuum seal **206** can be used to load gas into gas reservoir **208**. In some cases, the vacuum seal **206** can be used to shut off the connection to the gas handling system once the required amount of gas has been loaded into gas reservoir **208**.

(32) FIG. 3 is a diagram of a neutron generator **300** that includes an ion source **204**, a gas reservoir **208**, and a target **320**, in accordance with aspects of the present disclosure. In some instances, one or more of the components within neutron generator **300** may be coupled to one or more electrodes (e.g., electrodes **224**) that can be configured to provide biasing. For example, gas reservoir **208** may be coupled to an electrode corresponding to a gas reservoir voltage V.sub.R **324** and to an electrode that provides grounding (e.g., GND **326**).

(33) In some cases, the target **320** may include a film target (e.g., film target **210**) and a target rod (e.g., target rod **212**). In some examples, the ion source **204** may include a hot cathode **302**. In some configurations, hot cathode **302** may have a cylindrical form with a radius that is represented by r **312**. In some configurations, hot cathode **302** may be coupled to an electrode corresponding to a hot cathode voltage V.sub.HC **328** and to an electrode corresponding to ground (e.g., GND **326**). In some aspects, hot cathode **302** can be used to emit electrons (e.g., I.sub.e **318**) for direct ionization of gas molecules to produce D/T ions. In some configurations, ion beam **222** from the ion source **204** can be switched on/off by controlling the electron emission from the hot cathode **302** to stop ionization and by controlling the ion beam extraction inside the ion source **204**.

(34) In some examples, ion source **204** may include an ion source cylinder **304** that is associated with a first grid **306**. In some aspects, the first grid **306** may be coupled to an electrode that is configured to provide a grid voltage V.sub.G **330**. As illustrated, the distance d **314** represents the distance between the surface of the hot cathode **302** and the first grid **306**. In some configurations,

the ion source **204** may also include an extractor **308** with a second grid **310** that can be configured to extract the ion beam **222** from the ion source **204**. In some examples, the second grid may be coupled to an electrode that is configured to provide an extractor voltage $V_{\text{sub.E}}$ **332**. As illustrated, the length $L_{\text{sub.i}}$ **316** is a length between the first grid **306** and the extractor **308** (e.g., the second grid **310**) in the region of the ion source cylinder **304**, and the length $L_{\text{sub.a}}$ **322** is a length between the extractor **308** (e.g., the second grid **310**) and the target **320** in the acceleration volume. In some aspects, high voltages can be applied to the suppressor **214** (e.g., electrode corresponding to $V_{\text{sub.S}}$ **334**) and target **320** (e.g., electrode corresponding to $V_{\text{sub.T}}$ **336**) for accelerating the ion beam **222** to bombard the target **320**.

(35) In some aspects, hot cathode emission may be governed by the Child Law (or the Child-Langmuir Law or three-halves-power law). That is, the maximum space-charge-limited current in a planar diode structure can be a function of the distance and potential difference between the hot cathode **302** and the first grid **306**, provided that the hot cathode **302** is sufficiently heated such that sufficient electron charges hover near its surface space. In some cases, applying a given potential difference between the hot cathode **302** and the first grid **306** can cause the electron beam **318** to be extracted and shot, passing through the first grid **306**. In some cases, the first grid **306** can have a transparency that is approximately 90-100%. In some aspects, the electron current I_e can be represented as follows:

$$(36) \quad I_e = K \frac{V_g^{3/2}}{d^2} A = K \frac{V_g^{3/2}}{d^2} r^2 \quad (1)$$

(37) In equation (1), $I_{\text{sub.e}}$ can correspond to the electron current (mA); $V_{\text{sub.g}}$ can correspond to the voltage difference between cathode and grid (V); d can correspond to the distance between cathode and grid (mm); and A can correspond to the surface area of cathode with a radius of r (e.g., surface area may be measured in mm²). In some cases, for electrons, $k=0.002334 \text{ mA V}^{-3/2}$.

(38) FIGS. 2 and 3 illustrate neutron generators having a tube-like geometry, in which the hot cathode **302** is mounted and positioned next to the first grid **306** structure. As noted above, the radius of the hot cathode **302** surface is r **312**, the distance from the hot cathode **302** to the first grid **306** is d **314**, and the bias voltage on the first grid **306** is $V_{\text{sub.G}}$. In some aspects, hot cathode **302** can have a geometry similar to a tube-like structure. In one illustrative example, hot cathode **302** can have a radius $r=2 \text{ mm}$ and the distance from the hot cathode **302** to the first grid **306** can be approximately $d=1.0\text{-}1.5 \text{ mm}$. In some instances, equation (1) can be used to plot the electron current as a function of the bias voltage applied on the first grid **306**.

(39) In some cases, the hot cathode **302** can send an electron beam **318** to ionize hydrogen or hydrogen isotope (D/T) gas at a given pressure in the region of the ion source cylinder **304**. In some aspects, the ionized gas can be extracted (e.g., by the extractor **308**) in the form of an ion beam **222**. For hydrogen and hydrogen isotope molecular ionization, cross sections can be functions of electron impact energy in a range from 0 eV to a few keV. In some examples, an electron energy range of interest can be from 80 eV to 200 eV, while the cross sections can be in the range of 0.7 and 1.0 A_{sup.2} (an average of 0.85 A_{sup.2}), which may be equivalent to about one Bohr radius in size.

(40) In some configurations, assuming a close to 100% ion extraction efficiency, the ion beam **222** current extracted can be expressed in the following equation:

$$(41) \quad I_I = I_e n_{\text{DT}} L_I \quad (2)$$

(42) In equation (2), $I_{\text{sub.I}}$ can correspond to the ion current of the ion beam **222**; $I_{\text{sub.e}}$ can correspond to the electron current of the electron beam **318** from the hot cathode **302** (e.g., 50 mA); $L_{\text{sub.I}}$ can correspond to a length $L_{\text{sub.i}}$ **316** between the first grid **306** and the extractor **308** in the region of the ion source cylinder **304** (e.g., 1.0 cm); σ can correspond to the hydrogen molecular ionization cross section at a given electron energy (e.g., between 80-100 eV); and $n_{\text{sub.DT}}$ can correspond to the D-T molecular gas pressure in the region of the ion source cylinder **304** at a given heating power on the gas reservoir **208** (e.g., 1.0 mTorr).

(43) In one illustrative example that assumes the above operating parameters (e.g., 1.0 mTorr and 50 mA in a 1.0 cm geometry and assuming ~100% efficiency for ion extraction), 150 μ A ion current can pass therethrough. Both transparencies of the first grid **306** and the second grid **310** on the extractor **308** can reduce the final ion beam **222** current. In some cases, the gas pressure can be adjusted (e.g., increased) to compensate for ion losses.

(44) In some examples, the gas reservoir **208** can be heated to generate 1.0 mTorr or higher gas pressure. The hot cathode **302** is heated sufficiently so that a sufficient number of electron charges hover near its surface space. By applying a voltage (e.g., V.sub.G) on the first grid **306** in a range of 200-250 V, ion source **204** can generate an electron beam **318** with a current of 40-50 mA shooting into the region of the ion source cylinder **304**. With approximately 100-150 V in the middle region of the ion source cylinder **304**, the electrons will be deaccelerated for ionization with the highest cross-section to produce more ions. In some cases, ions can be extracted with the extractor **308** when a voltage is applied between 0 to -50 V.

(45) According to some examples, the ion source **204** can be based on electron-impact direct ionization. That is, in some instances, a plasma formation in the region of the ion source cylinder **304** may not be needed. Thus, the ion beam **222** may be pulsed at a relatively fast rate, with the pulse rise and fall times being in a range of 100-500 nsec. In some instances, the structure of the ion source **204** can have a small capacitance and impedance (no magnetic field) and the control voltages can be applied with values less than or equal to 300 V.

(46) According to some aspects, the capability of fast pulsing can make the neutron generator **300** useful for a variety of downhole measurements including fast neutron C/O—ratio of carbon and oxygen, and thermal neutron capture elemental analysis. Because of direct electron-impact ionization, the neutron generator **300** gas pressure can be a “free parameter” that can be used for adjusting the current of the ion beam **222**, along with the hot cathode **302** electron beam **318** emission. Thus, in a pulsed operation mode, the ion beam **222** current can be adjusted high, reversely proportional to the duty factor, to maintain a constant average ion beam current as if in a CW—continuous wave mode. The low gas pressure in the ion source **204**, combining with no real plasma formation, makes the pulsed operation much easy in control.

(47) FIG. 4A is a diagram of a hot cathode assembly **400** for use in an ion source of a neutron generator (e.g., ion source **204** of neutron generator **300**), in accordance with aspects of the present disclosure. In some cases, hot cathode assembly **400** can include hot cathode **401**. In some examples, hot cathode **401** can have a cylindrical shape. In some cases, the body of hot cathode **401** may include molybdenum-rhenium (MoRe). In some aspects, a first cylindrical sleeve such as mounting sleeve **402** can be coupled to hot cathode **401**. For instance, mounting sleeve **402** can be coupled to hot cathode **401** by welding, using an adhesive, using a mechanical component such as a clip or fastener, etc. In some cases, mounting sleeve **402** may be formed together with the hot cathode **401**.

(48) In some configurations, a second cylindrical sleeve such as thermal shielding sleeve **404** can be attached to mounting sleeve **402**. That is, the thermal shielding sleeve **404** may have a larger diameter than the mounting sleeve **402** to form a concentric configuration. In some instances, the thermal shielding sleeve **404** may be coupled to the mounting sleeve **402** using base member **406**. In some aspects, base member **406** may include a Kovar material, a Monel material, any other material suitable for coupling the cylindrical sleeves, and/or any combination thereof.

(49) In some examples, the mounting sleeve **402**, the thermal shielding sleeve **404**, and/or the base member **406** may be formed as part of a single case or housing. In further examples, one or more of the components illustrated (e.g., mounting sleeve **402**, thermal shielding sleeve **404**, base member **406**, etc.) may be coupled to hot cathode **401** as part of a manufacturing process and/or in a post-manufacturing assembly process.

(50) In some cases, thermal shielding sleeve **404** may include one or more openings such as vacuum through-holes **412**. In some examples, hot cathode **401** may include one or more electrical

contacts that can be used to provide biasing voltage(s) and/or grounding for hot cathode **401**. For example, hot cathode **401** can include electrical leg **410A** and electrical leg **410B**. In some configurations, electrical leg **410A** can be coupled (e.g., through a spot weld) to mounting sleeve **402**. That is, electrical leg **410A** can receive an electrical signal by way of mounting sleeve **402**, which can be coupled to an electrode within a neutron generator via thermal shielding sleeve **404** and mounting flange **408A**. In some examples, electrical leg **410B** may also be coupled to another electrode within a neutron generator (e.g., see FIG. 3).

(51) In some aspects, mounting flange **408A** can be attached to an outer sleeve (e.g., thermal shielding sleeve **404**). In some cases, mounting flange **408A** can be used to mount the hot cathode assembly **400** to the V.sub.HC electrode within a neutron generator. In some examples, the location that the mounting flange **408A** attaches to thermal shielding sleeve **404** can be used to determine the position of the hot cathode assembly **400** within the neutron generator. For example, as illustrated in FIG. 4A, mounting flange **408A** is attached to thermal shielding sleeve **404** at an intermediate location that substantially coincides with the middle of the body of hot cathode **401**, which can cause the body of hot cathode **401** to protrude past the mounting flange **408A** and the V.sub.HC electrode (see FIG. 5A). That is, the body of hot cathode **401** can extend past the V.sub.HC electrode by a distance that is based on the distance between the edge of the thermal shielding sleeve **404** and the connection point with mounting flange **408A**.

(52) FIG. 4B is a diagram of a hot cathode assembly **450** for use in an ion source of a neutron generator (e.g., ion source **204** of neutron generator **300**), in accordance with aspects of the present disclosure. In some cases, hot cathode assembly **450** is similar to hot cathode assembly **400**. That is, hot cathode assembly **450** includes a hot cathode **401**, mounting sleeve **402**, thermal shielding sleeve **404**, base member **406**, electrical leg **410A**, electrical leg **410B**, and/or vacuum through-holes **412**.

(53) In some aspects, hot cathode assembly **450** may include mounting flange **408B**. As noted above with respect to mounting flange **408A**, the location that the mounting flange **408B** attaches to thermal shielding sleeve **404** can be used to determine the position of the hot cathode assembly **450** within the neutron generator. As illustrated in FIG. 4B, mounting flange **408B** is attached to thermal shielding sleeve **404** at or near the end (e.g., edge) of the thermal shielding sleeve **404**, which can cause the body of hot cathode **401** to be aligned with the V.sub.HC electrode (see FIG. 5B).

(54) FIG. 5A is a diagram of a system **500** that includes a hot cathode assembly **400** configured within an ion source, in accordance with aspects of the present disclosure. In some examples, the hot cathode assembly illustrated in FIG. 5A can correspond to hot cathode assembly **400** and can include hot cathode **401**, mounting sleeve **402**, thermal shielding sleeve **404**, base member **406**, mounting flange **408A**, electrical leg **410A**, and electrical leg **410B**. In some aspects, mounting flange **408A** can be coupled to V.sub.HC electrode **502** (e.g., electrode that supplies bias voltage to hot cathode **401**). In some cases, electrical leg **410B** can be coupled to another electrode (not illustrated) that provides grounding to hot cathode **401**. In some cases, hot cathode **401** may share a common ground electrode with a gas reservoir (as illustrated in FIG. 6B).

(55) In some configurations, V.sub.HC electrode **502** can include a centralizing curb **508** that can be used to center or otherwise position the hot cathode assembly. For example, mounting flange **408A** may abut centralizing curb **508** to position the hot cathode assembly within the tube.

(56) In some cases, V.sub.HC electrode may also include vacuum through-holes **512** for vacuum connectivity.

(57) In some aspects, a portion of the body of hot cathode **401** may extend past the V.sub.HC electrode **502**. That is, as noted with respect to hot cathode assembly **400**, mounting flange **408A** can be coupled to an intermediate portion of thermal shielding sleeve **404** such that the end of hot cathode **401** is a distance d **506A** from the V.sub.G electrode **504**. In some cases, the V.sub.G electrode **504** may be aligned with a first grid (e.g., first grid **306**) that is associated with a source

cylinder.

(58) In some examples, V.sub.HC electrode **502** may include an e-field shaping structure **510** that shapes the electrical field in the gap between the V.sub.HC electrode **502** and the V.sub.G electrode **504**. In some aspects, the portion of thermal shielding sleeve **404** that extends past V.sub.HC electrode **502** may also be configured to operate as an e-field shaping structure.

(59) FIG. 5B is a diagram of a system **550** that includes a hot cathode assembly **450** configured within an ion source, in accordance with aspects of the present disclosure. In some examples, the hot cathode assembly illustrated in FIG. 5B can correspond to hot cathode assembly **450** and can include hot cathode **401**, mounting sleeve **402**, thermal shielding sleeve **404**, base member **406**, mounting flange **408B**, electrical leg **410A**, and electrical leg **410B**. In some aspects, mounting flange **408B** can be coupled to V.sub.HC electrode **502** (e.g., electrode that supplies bias voltage to hot cathode **401**). In some cases, electrical leg **410B** can be coupled to another electrode (not illustrated) that provides grounding to hot cathode **401**. In some cases, hot cathode **401** may share a common ground electrode with a gas reservoir (as illustrated in FIG. 6B).

(60) As noted with respect to FIG. 5A, V.sub.HC electrode **502** can include a centralizing curb **508** that can be used to center or otherwise position the hot cathode assembly. For example, mounting flange **408B** may abut centralizing curb **508** to position the hot cathode assembly within the tube.

(61) In some aspects, the end of hot cathode **401** may be aligned with the V.sub.HC electrode **502**. That is, as noted with respect to hot cathode assembly **450**, mounting flange **408B** can be coupled to an end portion of thermal shielding sleeve **404** such that the end of hot cathode **401** is substantially aligned with V.sub.HC electrode (e.g., a distance d **506B** from the V.sub.G electrode **504**).

(62) FIG. 6A is a diagram illustrating construction of a neutron generator **600** that includes a hot cathode assembly **400**. Note that FIG. 6A only illustrates the ion source side of the neutron generator **600** for simplicity (e.g., the suppressor and target are not illustrated). In some aspects, a first step for construction of neutron generator **600** may include brazing together the vacuum envelope of the tube, which may include ceramic spacers and electrodes (e.g., electrodes corresponding to V.sub.R **324**, GND **326**, V.sub.HC **328**, V.sub.G **330**, and V.sub.E **332**). In some aspects, neutron generator **600** may also include welding lip **601** that can be used for sealing the vacuum tube (e.g., with an end-cap **206**).

(63) In some examples, after the tube is brazed, construction may proceed by installing hot cathode assembly **400** into the tube of neutron generator **600**. In some aspects, hot cathode assembly **400** may be installed by coupling (e.g., welding, soldering, attaching) one or more components from hot cathode assembly **400** to electrodes of the tube. For instance, connection **602A** and connection **602B** illustrate coupling of mounting flange **408A** to an electrode corresponding to V.sub.HC **328**, and connection **604** illustrates coupling of electrical leg **410A** to an electrode corresponding to ground (e.g., GND **326**). As noted with respect to FIG. 5A and FIG. 5B, the electrode corresponding to V.sub.HC **328** may include a centralizing curb **508** that may abut mounting flange **408A**. Installation of hot cathode assembly **400** may be configured in accordance with a desired distance d **314** and/or in accordance with the dimensions of hot cathode **401** (e.g., according to radius r **312**).

(64) FIG. 6B is a diagram illustrating further steps for construction of a neutron generator **600** that includes a hot cathode assembly **400**. As noted with respect to FIG. 6A, FIG. 6B only illustrates the ion source side of the neutron generator **600** for simplicity. In some aspects, after hot cathode assembly **400** is installed, construction of neutron generator **600** may proceed by installing gas reservoir **208**. For example, connection **606** illustrates coupling of gas reservoir to an electrode corresponding to V.sub.R **324** and connection **608** illustrates coupling of gas reservoir to an electrode corresponding to ground (e.g., GND **326**). In some cases, gas reservoir **208** may be installed from the opening of vacuum tube envelope on the left side (e.g., prior to sealing the tube with end cap **206**).

(65) In some cases, after gas reservoir **208** is installed, construction of neutron generator **600** may proceed by installing (e.g., welding) an end cap **206** to seal the opening of vacuum tube envelope on the left side. For example, connection **610A** and connection **610B** illustrate coupling of the end cap **206** to the welding lip **601**. In some cases, the end cap **206** can have a tubing connection **612** to a vacuum pumping and gas handling system that can be used for processing before being pinched-off. In some aspects, construction of neutron generator **600** may include connecting (e.g., welding) the target (not illustrated) to seal the other end of the neutron tube.

(66) FIG. 7 is a graph of electron emission current as a function of grid voltage for a hot cathode of a neutron generator according to one example of the present disclosure. The chart shows electron emission current ($I_{\text{sub.e}}$) as a function of the voltage of the first grid **306** voltage (voltage of the first grid **306** minus the voltage of the hot cathode **302**) for a hot cathode **302** with a radius of 2 mm and positioned at 1.5 mm next to the first grid **306**, according to one example of the disclosure. With such a geometry, and the suitable hot cathode **302**, an electron current of 40-50 mA can be achieved when applying a voltage to the first grid **306** in a range of 200-250 V. Such a hot cathode can involve a few watts of heating power, and it can provide an operational lifetime beyond 10,000 hours. According to the voltage dependence of the Child Law, the higher the grid voltage is, the higher the electron emission current provided that sufficient space charge is available on the cathode surface due to heating. Therefore, with a separated grid, this can be optimized in the ion source by adjusting the voltage of the first grid higher, close to 250 V.

(67) FIG. 8 illustrates an example of a process **800** for constructing a neutron generator, in accordance with aspects of the present disclosure. At block **802**, the process **800** includes attaching a mounting sleeve to an exterior surface of a hot cathode. For example, mounting sleeve **402** can be attached to an exterior surface of hot cathode **401**.

(68) At block **804**, the process **800** includes attaching a thermal shielding sleeve to the mounting sleeve using at least one coupling base member, wherein the thermal shielding sleeve has a larger diameter than the mounting sleeve. For instance, thermal shielding sleeve **404** can be attached to mounting sleeve **402** using base member **406**.

(69) At block **806**, the process **800** includes attaching a proximal end of at least one mounting flange to the thermal shielding sleeve. For example, a first end of mounting flange **408A** can be attached to thermal shielding sleeve **404**.

(70) At block **808**, the process **800** includes attaching a distal end of the at least one mounting flange to a hot cathode electrode disposed on an interior portion of a neutron generating tube. For instance, a second end of mounting flange **408A** can be attached to V.sub.HC electrode **502**.

(71) In some cases, the hot cathode electrode includes at least one centralizing curb member, and wherein the at least one centralizing curb member abuts to the distal end of the at least one mounting flange. For example, V.sub.HC electrode **502** can include centralizing curb **508**, which can abut the distal end of mounting flange **408A**.

(72) In some examples, the proximal end of the at least one mounting flange is attached to an intermediate portion of the thermal shielding sleeve. For instance, the proximal end of mounting flange **408A** can be attached to an intermediate portion (e.g., an intermediate position) of thermal shielding sleeve **404**. In some aspects, the hot cathode electrode includes at least one field shaping member that extends in a direction orthogonal to the at least one mounting flange. For example, V.sub.HC electrode **502** can include e-field shaping structure **510** that extends in a direction orthogonal to mounting flange **408A**.

(73) In some cases, the proximal end of the at least one mounting flange is attached to an edge portion of the thermal shielding sleeve. For example, the proximal end of mounting flange **408B** can be attached to an edge portion (e.g., end) of thermal shielding sleeve **404**. In some instances, the mounting sleeve and the thermal shielding sleeve can comprise a molybdenum-rhenium (MoRe) material.

(74) In some examples, the process **800** can include brazing the stack of a vacuum envelope, which

may include ceramic rings and electrodes. For example, brazing the stack of the vacuum envelope corresponding to neutron generator **200**. In some cases, the process **800** can include installing (e.g., welding, attaching, etc.) the hot cathode assembly in a neutron generator (e.g., matching flange positions, making electrical connections to electrodes **224**, etc.). For example, mounting flange **408A** can be welded to V.sub.HC electrode **502**. In some instances, the process **800** can include installing the gas reservoir. For example, gas reservoir **208** can be installed (e.g., connected to electrodes) within neutron generator **200**. In some configurations, the process **800** can include welding the end cap **206** to the welding lip **601** to seal an end of the neutron tube, and the end cap **206** may include a tubing connection **612** to a vacuum pumping and filling system before being pinched-off. In some configurations, the process **800** can include installing (e.g., welding) the target to seal the opposite end of the neutron tube.

(75) FIG. **9** illustrates an example computing device architecture **900** which can be employed to perform various steps, methods, and techniques disclosed herein. Specifically, the techniques described herein can be implemented, at least in part, through the computing device architecture **900** in an applicable computing device, such computing device **126** and/or downhole tool **118**. The various implementations will be apparent to those of ordinary skill in the art when practicing the present technology. Persons of ordinary skill in the art will also readily appreciate that other system implementations or examples are possible.

(76) As noted above, FIG. **9** illustrates an example computing device architecture **900** of a computing device which can implement the various technologies and techniques described herein. The components of the computing device architecture **900** are shown in electrical communication with each other using a connection **905**, such as a bus. The example computing device architecture **900** includes a processing unit (CPU or processor) **910** and a computing device connection **905** that couples various computing device components including the computing device memory **915**, such as read only memory (ROM) **920** and random access memory (RAM) **925**, to the processor **910**.

(77) The computing device architecture **900** can include a cache of high-speed memory connected directly with, in close proximity to, or integrated as part of the processor **910**. The computing device architecture **900** can copy data from the memory **915** and/or the storage device **930** to the cache **912** for quick access by the processor **910**. In this way, the cache can provide a performance boost that avoids processor **910** delays while waiting for data. These and other modules can control or be configured to control the processor **910** to perform various actions. Other computing device memory **915** may be available for use as well. The memory **915** can include multiple different types of memory with different performance characteristics. The processor **910** can include any general purpose processor and a hardware or software service, such as service 1 **932**, service 2 **934**, and service 3 **936** stored in storage device **930**, configured to control the processor **910** as well as a special-purpose processor where software instructions are incorporated into the processor design. The processor **910** may be a self-contained system, containing multiple cores or processors, a bus, memory controller, cache, etc. A multi-core processor may be symmetric or asymmetric.

(78) To enable user interaction with the computing device architecture **900**, an input device **945** can represent any number of input mechanisms, such as a microphone for speech, a touch-sensitive screen for gesture or graphical input, keyboard, mouse, motion input, speech and so forth. An output device **935** can also be one or more of a number of output mechanisms known to those of skill in the art, such as a display, projector, television, speaker device, etc. In some instances, multimodal computing devices can enable a user to provide multiple types of input to communicate with the computing device architecture **900**. The communications interface **940** can generally govern and manage the user input and computing device output. There is no restriction on operating on any particular hardware arrangement and therefore the basic features here may easily be substituted for improved hardware or firmware arrangements as they are developed.

(79) Storage device **930** is a non-volatile memory and can be a hard disk or other types of computer readable media which can store data that are accessible by a computer, such as magnetic cassettes,

flash memory cards, solid state memory devices, digital versatile disks, cartridges, random access memories (RAMs) **925**, read only memory (ROM) **920**, and hybrids thereof. The storage device **930** can include services **932**, **934**, **936** for controlling the processor **910**. Other hardware or software modules are contemplated. The storage device **930** can be connected to the computing device connection **905**. In one aspect, a hardware module that performs a particular function can include the software component stored in a computer-readable medium in connection with the necessary hardware components, such as the processor **910**, connection **905**, output device **935**, and so forth, to carry out the function.

(80) For clarity of explanation, in some instances the present technology may be presented as including individual functional blocks including functional blocks comprising devices, device components, steps or routines in a method embodied in software, or combinations of hardware and software.

(81) In some embodiments the computer-readable storage devices, mediums, and memories can include a cable or wireless signal containing a bit stream and the like. However, when mentioned, non-transitory computer-readable storage media expressly exclude media such as energy, carrier signals, electromagnetic waves, and signals per se.

(82) Methods according to the above-described examples can be implemented using computer-executable instructions that are stored or otherwise available from computer readable media. Such instructions can include, for example, instructions and data which cause or otherwise configure a general purpose computer, special purpose computer, or a processing device to perform a certain function or group of functions. Portions of computer resources used can be accessible over a network. The computer executable instructions may be, for example, binaries, intermediate format instructions such as assembly language, firmware, source code, etc. Examples of computer-readable media that may be used to store instructions, information used, and/or information created during methods according to described examples include magnetic or optical disks, flash memory, USB devices provided with non-volatile memory, networked storage devices, and so on.

(83) Devices implementing methods according to these disclosures can include hardware, firmware and/or software, and can take any of a variety of form factors. Typical examples of such form factors include laptops, smart phones, small form factor personal computers, personal digital assistants, rackmount devices, standalone devices, and so on. Functionality described herein also can be embodied in peripherals or add-in cards. Such functionality can also be implemented on a circuit board among different chips or different processes executing in a single device, by way of further example.

(84) The instructions, media for conveying such instructions, computing resources for executing them, and other structures for supporting such computing resources are example means for providing the functions described in the disclosure.

(85) In the foregoing description, aspects of the application are described with reference to specific embodiments thereof, but those skilled in the art will recognize that the application is not limited thereto. Thus, while illustrative embodiments of the application have been described in detail herein, it is to be understood that the disclosed concepts may be otherwise variously embodied and employed, and that the appended claims are intended to include such variations, except as limited by the prior art. Various features and aspects of the above-described subject matter may be used individually or jointly. Further, embodiments can be utilized in any number of environments and applications beyond those described herein without departing from the broader spirit and scope of the specification. The specification and drawings are, accordingly, to be regarded as illustrative rather than restrictive. For the purposes of illustration, methods were described in a particular order. It should be appreciated that in alternate embodiments, the methods may be performed in a different order than that described.

(86) Where components are described as being “configured to” perform certain operations, such configuration can be accomplished, for example, by designing electronic circuits or other hardware

to perform the operation, by programming programmable electronic circuits (e.g., microprocessors, or other suitable electronic circuits) to perform the operation, or any combination thereof.

(87) The various illustrative logical blocks, modules, circuits, and algorithm steps described in connection with the examples disclosed herein may be implemented as electronic hardware, computer software, firmware, or combinations thereof. To clearly illustrate this interchangeability of hardware and software, various illustrative components, blocks, modules, circuits, and steps have been described above generally in terms of their functionality. Whether such functionality is implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system. Skilled artisans may implement the described functionality in varying ways for each particular application, but such implementation decisions should not be interpreted as causing a departure from the scope of the present application.

(88) The techniques described herein may also be implemented in electronic hardware, computer software, firmware, or any combination thereof. Such techniques may be implemented in any of a variety of devices such as general purposes computers, wireless communication device handsets, or integrated circuit devices having multiple uses including application in wireless communication device handsets and other devices. Any features described as modules or components may be implemented together in an integrated logic device or separately as discrete but interoperable logic devices. If implemented in software, the techniques may be realized at least in part by a computer-readable data storage medium comprising program code including instructions that, when executed, performs one or more of the method, algorithms, and/or operations described above. The computer-readable data storage medium may form part of a computer program product, which may include packaging materials.

(89) The computer-readable medium may include memory or data storage media, such as random access memory (RAM) such as synchronous dynamic random access memory (SDRAM), read-only memory (ROM), non-volatile random access memory (NVRAM), electrically erasable programmable read-only memory (EEPROM), FLASH memory, magnetic or optical data storage media, and the like. The techniques additionally, or alternatively, may be realized at least in part by a computer-readable communication medium that carries or communicates program code in the form of instructions or data structures and that can be accessed, read, and/or executed by a computer, such as propagated signals or waves.

(90) Other embodiments of the disclosure may be practiced in network computing environments with many types of computer system configurations, including personal computers, hand-held devices, multi-processor systems, microprocessor-based or programmable consumer electronics, network PCs, minicomputers, mainframe computers, and the like. Embodiments may also be practiced in distributed computing environments where tasks are performed by local and remote processing devices that are linked (either by hardwired links, wireless links, or by a combination thereof) through a communications network. In a distributed computing environment, program modules may be located in both local and remote memory storage devices.

(91) In the above description, terms such as “upper,” “upward,” “lower,” “downward,” “above,” “below,” “downhole,” “uphole,” “longitudinal,” “lateral,” and the like, as used herein, shall mean in relation to the bottom or furthest extent of the surrounding wellbore even though the wellbore or portions of it may be deviated or horizontal. Correspondingly, the transverse, axial, lateral, longitudinal, radial, etc., orientations shall mean orientations relative to the orientation of the wellbore or tool.

(92) The term “coupled” is defined as connected, whether directly or indirectly through intervening components, and is not necessarily limited to physical connections. The connection can be such that the objects are permanently connected or releasably connected. The term “outside” refers to a region that is beyond the outermost confines of a physical object. The term “inside” indicates that at least a portion of a region is partially contained within a boundary formed by the object. The term “substantially” is defined to be essentially conforming to the particular dimension, shape or

another word that substantially modifies, such that the component need not be exact. For example, substantially cylindrical means that the object resembles a cylinder, but can have one or more deviations from a true cylinder.

(93) The term “radially” means substantially in a direction along a radius of the object, or having a directional component in a direction along a radius of the object, even if the object is not exactly circular or cylindrical. The term “axially” means substantially along a direction of the axis of the object. If not specified, the term axially is such that it refers to the longer axis of the object.

(94) Although a variety of information was used to explain aspects within the scope of the appended claims, no limitation of the claims should be implied based on particular features or arrangements, as one of ordinary skill would be able to derive a wide variety of implementations. Further and although some subject matter may have been described in language specific to structural features and/or method steps, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to these described features or acts. Such functionality can be distributed differently or performed in components other than those identified herein. The described features and steps are disclosed as possible components of systems and methods within the scope of the appended claims.

(95) Moreover, claim language reciting “at least one of” a set indicates that one member of the set or multiple members of the set satisfy the claim. For example, claim language reciting “at least one of A and B” means A, B, or A and B.

(96) Statements of the disclosure include:

(97) Statement 1. A neutron generator comprising: an ion source for generating neutrons for neutron logging downhole in a wellbore, wherein the ion source includes: a hot cathode; at least one cylindrical sleeve coupled to an exterior surface of the hot cathode; at least one electrode coupled to an interior surface of the neutron generator; and at least one mounting flange, wherein a proximal end of the at least one mounting flange is coupled to the at least one cylindrical sleeve and a distal end of the at least one mounting flange is coupled to the at least one electrode.

(98) Statement 2. The neutron generator of statement 1, wherein the at least one cylindrical sleeve comprises a first cylindrical sleeve and a second cylindrical sleeve that is concentric to the first cylindrical sleeve, wherein the first cylindrical sleeve is coupled to the exterior surface of the hot cathode and the second cylindrical sleeve is coupled to the proximal end of the at least one mounting flange.

(99) Statement 3. The neutron generator of statement 2, further comprising: a base member having a first end that is coupled to the first cylindrical sleeve and a second end that is coupled to the second cylindrical sleeve.

(100) Statement 4. The neutron generator of any of statements 1 to 3, further comprising: at least one centralizing curb member protruding from the at least one electrode, wherein the at least one centralizing curb member abuts to the distal end of the at least one mounting flange.

(101) Statement 5. The neutron generator of any of statements 1 to 4, wherein the proximal end of the at least one mounting flange is coupled to an intermediate portion of the at least one cylindrical sleeve, wherein a portion of the hot cathode extends past the at least one electrode.

(102) Statement 6. The neutron generator of any of statements 1 to 4, wherein the proximal end of the at least one mounting flange is coupled to an edge portion of the at least one cylindrical sleeve, wherein an end of the hot cathode is aligned with the at least one electrode.

(103) Statement 7. The neutron generator of any of statements 1 to 6, wherein the at least one electrode corresponds to a hot cathode biasing voltage.

(104) Statement 8. A hot cathode apparatus comprising: a hot cathode; at least one cylindrical sleeve coupled to an exterior surface of the hot cathode; and at least one mounting flange, wherein a proximal end of the at least one mounting flange is coupled to the at least one cylindrical sleeve and a distal end of the at least one mounting flange is configured for attaching to at least one electrode for providing a bias voltage to the hot cathode.

(105) Statement 9. The hot cathode apparatus of statement 8, wherein the at least one cylindrical sleeve comprises a first cylindrical sleeve and a second cylindrical sleeve that is concentric to the first cylindrical sleeve, wherein the first cylindrical sleeve is coupled to the exterior surface of the hot cathode and the second cylindrical sleeve is coupled to the proximal end of the at least one mounting flange.

(106) Statement 10. The hot cathode apparatus of statement 9, further comprising: a base member having a first end that is coupled to the first cylindrical sleeve and a second end that is coupled to the second cylindrical sleeve.

(107) Statement 11. The hot cathode apparatus of any of statements 9 to 10, wherein the second cylindrical sleeve includes a plurality of apertures for vacuum connectivity.

(108) Statement 12. The hot cathode apparatus of any of statements 8 to 11, wherein the at least one cylindrical sleeve comprises a molybdenum-rhenium (MoRe) material.

(109) Statement 13. The hot cathode apparatus of any of statements 8 to 12, wherein the proximal end of the at least one mounting flange is coupled to an intermediate portion of the at least one cylindrical sleeve.

(110) Statement 14. The hot cathode apparatus of any of statements 8 to 12, wherein the proximal end of the at least one mounting flange is coupled to an edge portion of the at least one cylindrical sleeve.

(111) Statement 15. A method for constructing a neutron generator, the method comprising: attaching a mounting sleeve to an exterior surface of a hot cathode; attaching a thermal shielding sleeve to the mounting sleeve using at least one coupling base member, wherein the thermal shielding sleeve has a larger diameter than the mounting sleeve; attaching a proximal end of at least one mounting flange to the thermal shielding sleeve; and attaching a distal end of the at least one mounting flange to a hot cathode electrode disposed on an interior portion of a neutron generating tube.

(112) Statement 16. The method of statement 15, wherein the hot cathode electrode includes at least one centralizing curb member, and wherein the at least one centralizing curb member abuts to the distal end of the at least one mounting flange.

(113) Statement 17. The method of any of statements 15 to 16, wherein the proximal end of the at least one mounting flange is attached to an intermediate portion of the thermal shielding sleeve.

(114) Statement 18. The method of statement 17, wherein the hot cathode electrode includes at least one field shaping member that extends in a direction orthogonal to the at least one mounting flange.

(115) Statement 19. The method of any of statements 15 to 16, wherein the proximal end of the at least one mounting flange is attached to an edge portion of the thermal shielding sleeve.

(116) Statement 20. The method of any of statements 15 to 19, wherein the mounting sleeve and the thermal shielding sleeve comprise a molybdenum-rhenium (MoRe) material.

(117) Statement 21. An apparatus comprising at least one memory; and at least one processor coupled to the at least one memory, wherein the at least one processor is configured to perform operations in accordance with any one of statements 15 to 19.

(118) Statement 22: An apparatus comprising means for performing operations in accordance with any one of statements 15 to 19.

(119) Statement 23: A non-transitory computer-readable medium comprising instructions that, when executed by an apparatus, cause the apparatus to perform operations in accordance with any one of statements 15 to 19.

(120) Statement 24. A method comprising: deploying a logging tool having a neutron generator into a wellbore, the neutron generator comprising an ion source that includes: a hot cathode, at least one cylindrical sleeve coupled to an exterior surface of the hot cathode, at least one electrode coupled to an interior surface of the neutron generator, and at least one mounting flange, wherein a proximal end of the at least one mounting flange is coupled to the at least one cylindrical sleeve

and a distal end of the at least one mounting flange is coupled to the at least one electrode; ionizing an ionizable gas within the ion source to create a plurality of ions; accelerating the plurality of ions toward a target to yield a plurality of neutrons; transmitting the plurality of neutrons from the neutron generator into a formation surrounding the wellbore; and receiving a signal measurement related to the plurality of neutrons at one or more sensors in the logging tool.

(121) Statement 25. An apparatus comprising at least one memory; and at least one processor coupled to the at least one memory, wherein the at least one processor is configured to perform operations in accordance with statement 24.

(122) Statement 26: An apparatus comprising means for performing operations in accordance with statement 24.

(123) Statement 27: A non-transitory computer-readable medium comprising instructions that, when executed by an apparatus, cause the apparatus to perform operations in accordance with statement 24.

Claims

1. A neutron generator comprising: an ion source for generating neutrons for neutron logging downhole in a wellbore, wherein the ion source includes: a hot cathode; at least one cylindrical sleeve coupled to an exterior surface of the hot cathode; at least one electrode coupled to an interior surface of the neutron generator; and at least one mounting flange, wherein a proximal end of the at least one mounting flange is coupled to the at least one cylindrical sleeve and a distal end of the at least one mounting flange is coupled to the at least one electrode, further wherein the at least one cylindrical sleeve comprises a first cylindrical sleeve and a second cylindrical sleeve that is concentric to the first cylindrical sleeve, wherein the first cylindrical sleeve is coupled to the exterior surface of the hot cathode and the second cylindrical sleeve is coupled to the proximal end of the at least one mounting flange.

2. The neutron generator of claim 1, further comprising: a base member having a first end that is coupled to the first cylindrical sleeve and a second end that is coupled to the second cylindrical sleeve.

3. The neutron generator of claim 1, further comprising: at least one centralizing curb member protruding from the at least one electrode, wherein the at least one centralizing curb member abuts to the distal end of the at least one mounting flange.

4. The neutron generator of claim 1, wherein the proximal end of the at least one mounting flange is coupled to an intermediate portion of the at least one cylindrical sleeve, wherein a portion of the hot cathode extends past the at least one electrode.

5. The neutron generator of claim 1, wherein the proximal end of the at least one mounting flange is coupled to an edge portion of the at least one cylindrical sleeve, wherein an end of the hot cathode is aligned with the at least one electrode.

6. The neutron generator of claim 1, wherein the at least one electrode corresponds to a hot cathode biasing voltage.

7. A hot cathode apparatus comprising: a hot cathode; at least one cylindrical sleeve coupled to an exterior surface of the hot cathode; and at least one mounting flange, wherein a proximal end of the at least one mounting flange is coupled to the at least one cylindrical sleeve and a distal end of the at least one mounting flange is configured for attaching to at least one electrode for providing a bias voltage to the hot cathode, further wherein the at least one cylindrical sleeve comprises a first cylindrical sleeve and a second cylindrical sleeve that is concentric to the first cylindrical sleeve, wherein the first cylindrical sleeve is coupled to the exterior surface of the hot cathode and the second cylindrical sleeve is coupled to the proximal end of the at least one mounting flange.

8. The hot cathode apparatus of claim 7, further comprising: a base member having a first end that is coupled to the first cylindrical sleeve and a second end that is coupled to the second cylindrical

sleeve.

9. The hot cathode apparatus of claim 7, wherein the second cylindrical sleeve includes a plurality of apertures for vacuum connectivity.

10. The hot cathode apparatus of claim 7, wherein the at least one cylindrical sleeve comprises a molybdenum-rhenium (MoRe) material.

11. The hot cathode apparatus of claim 7, wherein the proximal end of the at least one mounting flange is coupled to an intermediate portion of the at least one cylindrical sleeve.

12. The hot cathode apparatus of claim 7, wherein the proximal end of the at least one mounting flange is coupled to an edge portion of the at least one cylindrical sleeve.

13. A method for constructing a neutron generator, the method comprising: attaching a mounting sleeve to an exterior surface of a hot cathode; attaching a thermal shielding sleeve to the mounting sleeve using at least one coupling base member, wherein the thermal shielding sleeve has a larger diameter than the mounting sleeve; attaching a proximal end of at least one mounting flange to the thermal shielding sleeve; and attaching a distal end of the at least one mounting flange to a hot cathode electrode disposed on an interior portion of a neutron generating tube.

14. The method of claim 13, wherein the hot cathode electrode includes at least one centralizing curb member, and wherein the at least one centralizing curb member abuts to the distal end of the at least one mounting flange.

15. The method of claim 13, wherein the proximal end of the at least one mounting flange is attached to an intermediate portion of the thermal shielding sleeve.

16. The method of claim 15, wherein the hot cathode electrode includes at least one field shaping member that extends in a direction orthogonal to the at least one mounting flange.

17. The method of claim 13, wherein the proximal end of the at least one mounting flange is attached to an edge portion of the thermal shielding sleeve.

18. The method of claim 13, wherein the mounting sleeve and the thermal shielding sleeve comprise a molybdenum-rhenium (MoRe) material.
